



Dropbox Shuts Down Mailbox

Dropbox, which acquired Mailbox for close to \$100 million in 2013, has announced that it is shutting down for good. <http://dgit.in/ByeMailbox>



Firefox OS

RIP Firefox OS

Mozilla announced that it will no longer sell or develop Firefox OS smart phones or devices. <http://dgit.in/RIPfirefoxOS>

Powering the Oculus, cleaning the cabinet and silly phone problems

Our readers are at it again and we're here to solve their peculiar problems

Configuration for Oculus Rift

I need some help to purchase a high end customized computer for my college lab. I work in my college and the basic requirement is to run a 3D driving simulator. The simulator is programmed in Java and uses JMonkeyEngine (JRE8). As per my research, the requirement for the simulator to provide the for best performance are:

- 1) Processor > 3 GHz
- 2) 8 GB Memory
- 3) NVIDIA GTX 970 or better

Aside from these, here are some considerations that need to be taken into account.

- 1) Cost should be less than ₹50,000/-
- 2) HDMI port & 64-bit support (for Oculus Rift DK2)

I have a few other related queries:

- 1) Which one would be best keeping in mind the requirements, AMD or Intel processors?
- 2) Are there any other cheaper equivalent to the Nvidia GTX 970?
- 3) What else would be required to meet the above said specification, say power supply(600-750 W), cooling systems, cabinet?

Awaiting your reply, as I have to forward the procurement draft to the college authorities.

—Vipul Nair

A Since you wish to run the simulator on the Oculus Rift, we'll read up on Oculus' official recommendations noted on a blog post highlighting its hardware requirements. (<http://dgit.in/RiftDK2HW>) So you'll need atleast:

- Intel Core i5 - 4590 or better
- GTX 970 or better
- 8 GB RAM or more

The processor costs ₹16.4K but for a little less you can get a newer gen CPU - Intel Core i5 - 6500 for ₹15.8K. To balance out the cost, the cheapest compatible motherboard we could find online

was the ASUS B150M-A D3 for ₹6,868. Likewise, for the RAM we found the Crucial PC4-1700 that cost ₹2,500, so two modules would cost you ₹5,000. As for the video card, the ZOTAC GeForce GTX 970 ZT-90101-10P costs ₹25,950 which is quite a good deal but it is likely that these components might not be the cheapest options by the time you search for them so check out a few price comparison websites before making the purchase. These newer generation of CPUs are quite cool so you can make do with a stock cooler but if you were to move to code with higher computational workload then an



Powering the Oculus

aftermarket cooler like the CoolerMaster TX3 would be an economical and prudent choice. Throw in one or two extra 120mm fans and you should be set. Your power requirements with the above configuration is around 400 W. So by incorporating a little headroom, any 550 W power supply should be enough. The Corsair VS550 is a pretty decent power supply for ₹3,100. A Cabinet hasn't been accounted for, you can pick any cabinet that has enough clearance for the graphics card. And we're assuming you guys already have hard drives.

The cost for these comes down to a little over ₹57,000 (without HDD/Cabinet/Monitor).

As for the other queries:

1. If you are looking at an equivalent in terms of benchmark scores, the

AMD FX 8320 and FX 8350 which cost roughly 12K and 14.5K will do the job. But here's something interesting to note, AMD came out with Project Quantum which is a pre-built machine for VR units using AMD's Fiji graphics card, however, this uses an Intel processor. This is likely because there are certain instruction sets that Intel processors are better built to handle. Plus the savings aren't that great, so unless a side by side comparison is done, it'd be wise to go with the configuration that Oculus has recommended.

2. The Oculus Rift has two screens, each being of 1080 x 1200 resolution/pixel dimension. Higher resolution requires more memory. GTX 960 is designed to optimally utilise 2GB, though 4GB variants exist now. The GTX 970 is designed for multi-display setups with increased bandwidth to push more data. And it's not just the resolution. You even require a frame rate of 75 FPS or more to reduce the chances of getting nasty headaches with the Oculus. 75 FPS is a high frame rate and your graphics card needs to handle not one but two displays at the same time. While the JMonkey Engine simulator will not be resource heavy and a GTX 960 should easily be able to manage the load, we're not sure since we haven't benchmarked an Oculus Rift. So if you can manage to get a GTX 960 from someone for a short period of time you might be able to run the simulation on two screens to check if the display is smooth (75 FPS or more).

Doing the Shuffle

I have an Apple iPod Shuffle. It's quite old so it works only with Windows XP. Recently, my desktop computer had some trouble in starting up. The CPU flashes the power